Exploratory Data Analysis Report

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1. Executive Summary

This report documents the Exploratory Data Analysis (EDA) performed on the Online Retail II dataset. The analysis spans eight structured sections: descriptive statistics, revenue trend analysis, product analysis, customer analysis, geographic analysis, correlation analysis, anomaly detection, and a composite summary dashboard. Each section is supported by visualisations generated from the cleaned dataset.

The dataset covers transactional records from a UK-based online retailer between December 2010 and December 2011 — over 333,000 transactions across 37 countries and 4,300+ unique customers. Key findings are summarised below:

- Revenue peaked at £642,985 in November 2011, confirming a strong Q4 seasonal demand cycle.
- The top 20% of customers generate approximately 65% of total revenue — consistent with the Pareto principle.
- The United Kingdom accounts for 85.6% of total revenue; Germany, France, and EIRE are the highest-potential international markets.
- Thursday at 12:00 noon is the peak trading window; Saturday registers near-zero activity, consistent with B2B wholesale operations.
- All three key numerical variables (Quantity, Price, Total_Revenue) exhibit right-skewed distributions with high kurtosis.
- Quantity and revenue show a moderate positive correlation ($r = 0.513$); month has negligible correlation with revenue ($r = -0.027$).
- The £1–3 price band drives the highest transaction volume and total revenue of any price segment.

2. Dataset Overview

The dataset used in this EDA was produced following the ETL pipeline stage and contains the following structure:

Table 1- Dataset Overview

Total Transactions	333,000+
Unique Customers	4,300+

Unique Products	3,800+ SKUs
Countries	37
Date Range	December 2010 – December 2011
Key Features	Quantity, Price, Total_Revenue
Engineered Features	DayOfWeek, Hour, Month, Year, Price_Band
Target for Modelling	Revenue, Customer Segments, Demand Trends

3. Descriptive Statistics

3.1 Numerical Variable Distributions

Figure 1 presents histograms and box plots for the three primary numerical variables: Quantity, Price, and Total_Revenue. The dashed red line in each histogram marks the mean value.

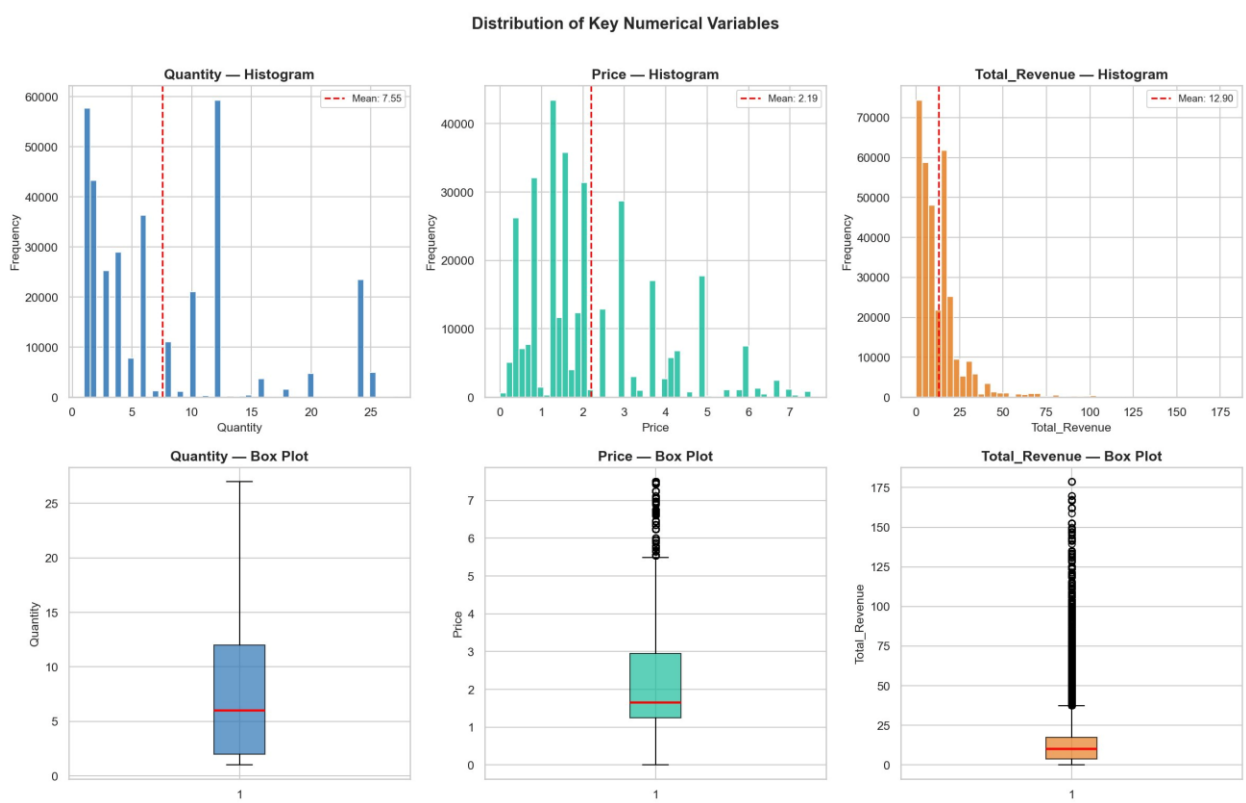

Figure 1 Distribution of Key Numerical Variables — Histograms and Box Plots (Quantity, Price, Total Revenue)

All three variables display pronounced right skew: the majority of transactions involve small quantities at low unit prices, while a small number of bulk or high-value orders create extended right tails. This pattern is characteristic of a B2B/wholesale retail context.

Table 2 - Mean ,Median, Skew

Variable	Mean	Median	Skew
Quantity	7.55 units	6	Positive
Price (£)	£2.19	£1.65	Positive
Total Revenue	£12.90	£9.90	Positive

The high coefficient of variation for all three variables confirms substantial spread around the mean. The leptokurtic (heavy-tailed) distribution of Total_Revenue indicates the presence of outlying high-value transactions that warrant further anomaly investigation (see Section 9).

4. Revenue Trend Analysis

4.1 Monthly Revenue Trend

Figure 2 plots total monthly revenue across the dataset period alongside a 3-month rolling average. The rolling average smooths short-term fluctuations to reveal the underlying structural trend.

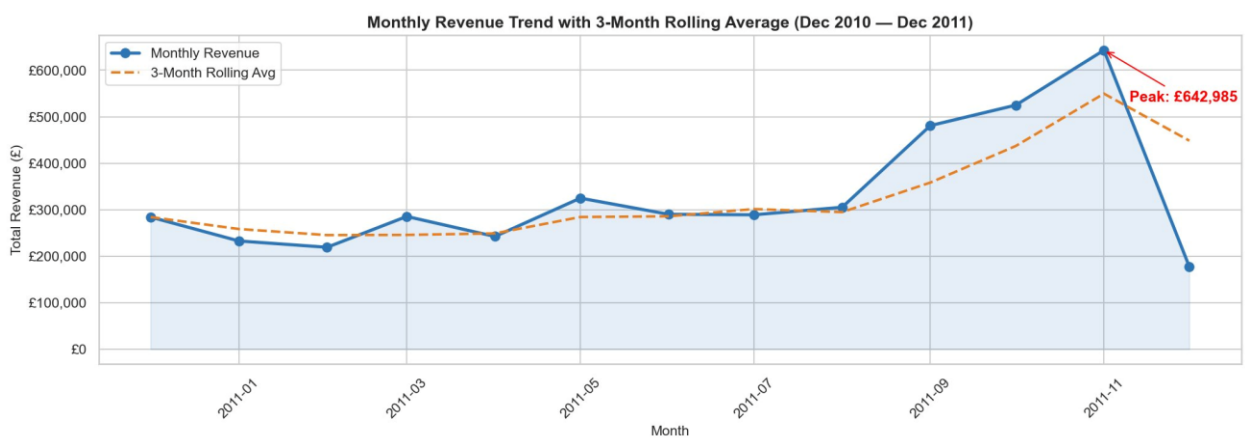

Figure 2 -Monthly Revenue Trend with 3-Month Rolling Average (Dec 2010 – Dec 2011)

Peak Revenue: November 2011

November 2011 recorded the highest monthly revenue at £642,985. The rolling average confirms this is the apex of a sustained Q4 growth phase beginning in September.

- Revenue remained relatively flat between January and August 2011 (approximately £230,000–£330,000/month).
- A sustained upward shift begins from September 2011, consistent with pre-Christmas wholesale stocking patterns.
- The December 2011 dip to ~£175,000 reflects a partial month of data rather than a structural decline.

4.2 Monthly Order Volume vs Revenue

Figure 3 overlays order count (bar chart) and revenue (line chart) on a dual-axis plot.

This comparison reveals whether Q4 growth is driven by more orders, higher average order values, or both.

Figure 3 - Monthly Order Volume vs Revenue — Dual-Axis Comparison

Order volume and revenue move together throughout the year. The November 2011 peak sees approximately 2,400 orders alongside peak revenue, confirming that Q4 growth is driven by both higher order frequency and elevated average order values. This dual-driver pattern is important for forecasting, as demand planning must account for both volume and value uplift.

4.3 Temporal Demand Patterns

Figure 4 disaggregates revenue by day of the week and hour of the day, revealing intra-week and intra-day demand cycles.

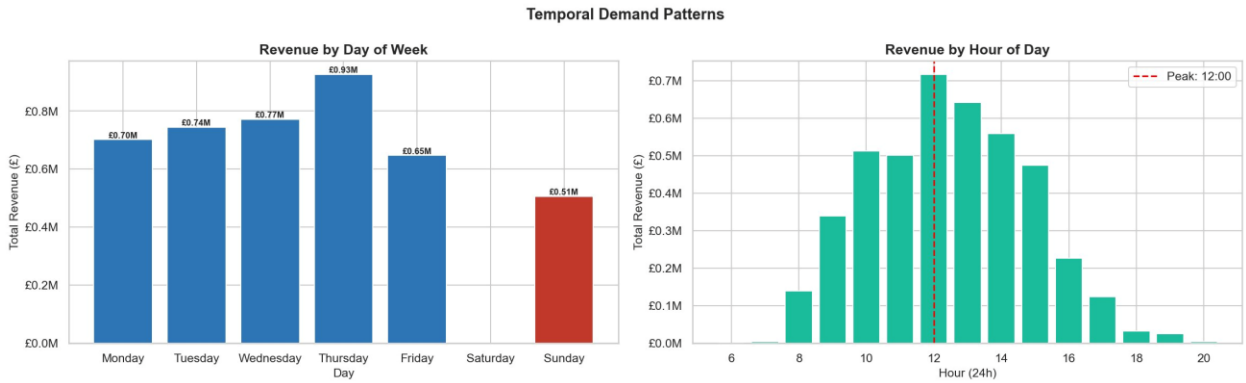

Figure 4 - Temporal Demand Patterns — Revenue by Day of Week and Hour of Day

Table 3-— Data by Revenue by Day of Week and Hour of Day

Peak Day	Thursday — £0.93M total revenue
Zero Activity	Saturday — near-zero revenue across the full dataset
Peak Hour	12:00 noon — highest single-hour revenue
Active Window	08:00–16:00 Monday through Friday
Interpretation	B2B wholesale model — business customers order within business hours

Saturday's near-zero revenue is a strong structural signal of a B2B wholesale customer base. Unlike B2C retailers where weekend demand is typically high, this dataset shows a clear five-day working-week pattern. This has implications for logistics, staffing, and marketing scheduling decisions.

5. Product Analysis

5.1 Top 10 Products by Revenue

Figure 5 ranks the top 10 products by total revenue generated over the full dataset period.

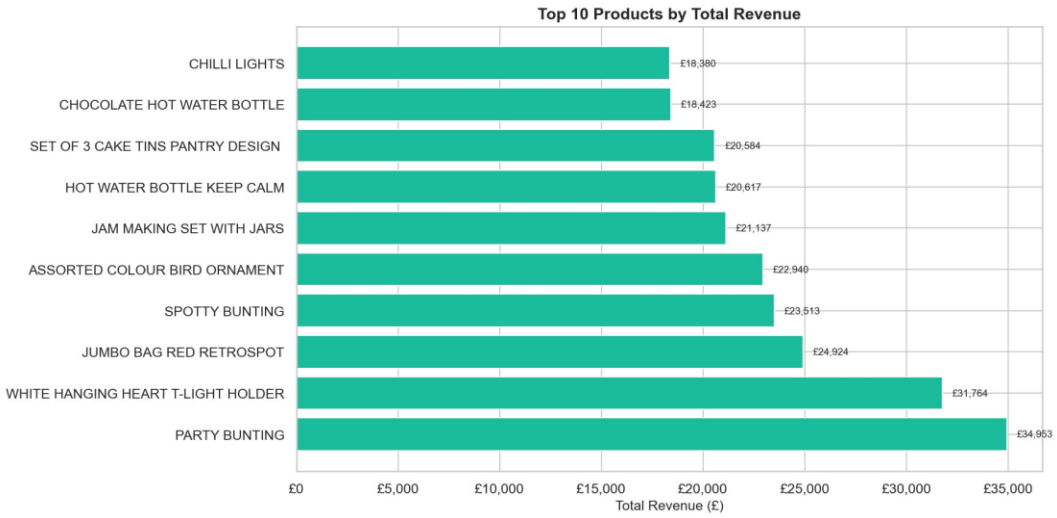

Figure 5 - Top 10 Products by Total Revenue (£)

PARTY BUNTING leads all products with £34,953 in revenue, followed by WHITE HANGING HEART T-LIGHT HOLDER at £31,764. These are mid-to-high price point products that generate strong revenue through moderate volumes. The revenue-leading products are predominantly decorative and gift items — consistent with the retailer's seasonal gifting focus.

5.2 Top 10 Products by Units Sold

Figure 6 ranks the top 10 products by total units sold. This differs meaningfully from the revenue ranking.

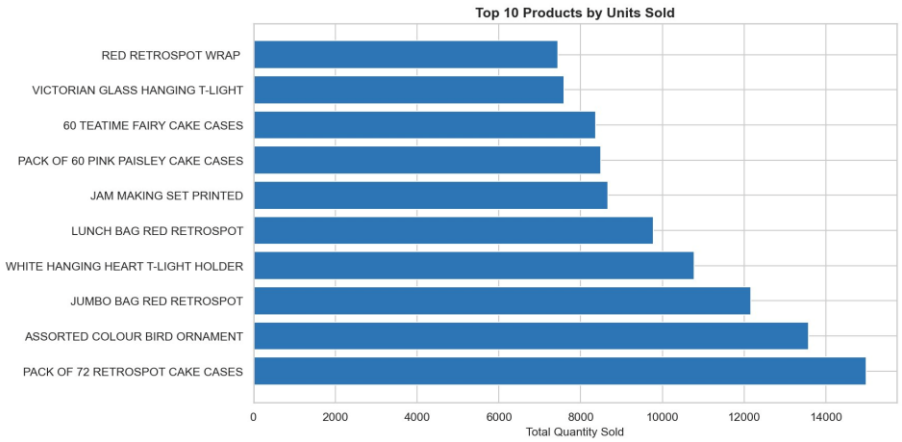

Figure 6 - Top 10 Products by Total Units Sold

PACK OF 72 RETROSPOT CAKE CASES leads by volume at approximately 15,000 units — nearly double the next-highest product. High-volume products are typically low-unit-price consumables (cake cases, wrapping, bags), while high-revenue products are higher-priced decorative items. This bifurcation has direct implications for inventory planning: volume leaders require high stock depth, while revenue leaders require margin protection and promotion.

5.3 Price Band Segmentation

Figure 7 segments all transactions into five price bands and shows both transaction count and total revenue contribution for each band.

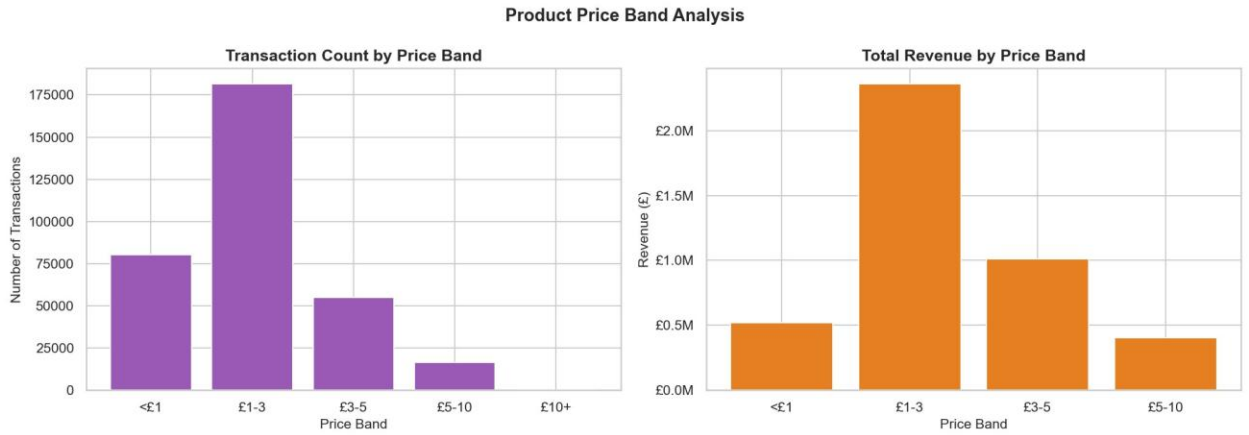

Figure 7 - Product Price Band Analysis — Transaction Count (left) and Revenue (right) by Price Band

£1–3 Price Band Dominates

The £1–3 band accounts for over 175,000 transactions and approximately £2.0M+ in total revenue — the highest of any band across both dimensions. This is the operational core of the product catalogue.

- Products priced below £1 generate high volume but low revenue contribution.
- The £10+ band is nearly absent from the dataset, confirming the retailer operates in the low-to-mid price range.
- The £3–5 band is the second largest revenue contributor, representing a key secondary segment.

6. Customer Analysis

6.1 Top 20 Customers by Revenue

Figure 8 ranks the top 20 customers by total revenue. The concentration visible in this chart is a key analytical finding.

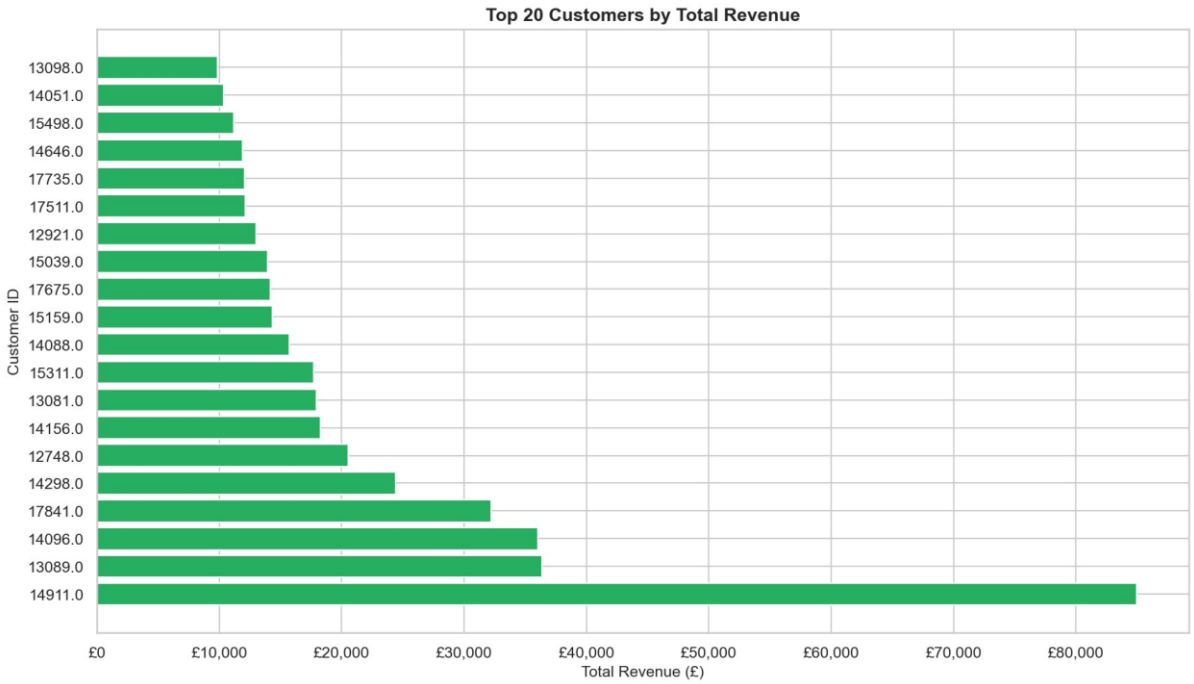

Figure 8 -Top 20 Customers by Total Revenue (£)

Customer 14911 — Outlier Account

Customer 14911 generated over £85,000 in total revenue — more than double the second-highest customer (14096/13089 at ~£35,000). This single account represents an extraordinary concentration of value and demands dedicated relationship management.

6.2 Customer Purchase Frequency

Figure 9 shows the distribution of orders per customer, presented as both a histogram and a segmented pie chart.

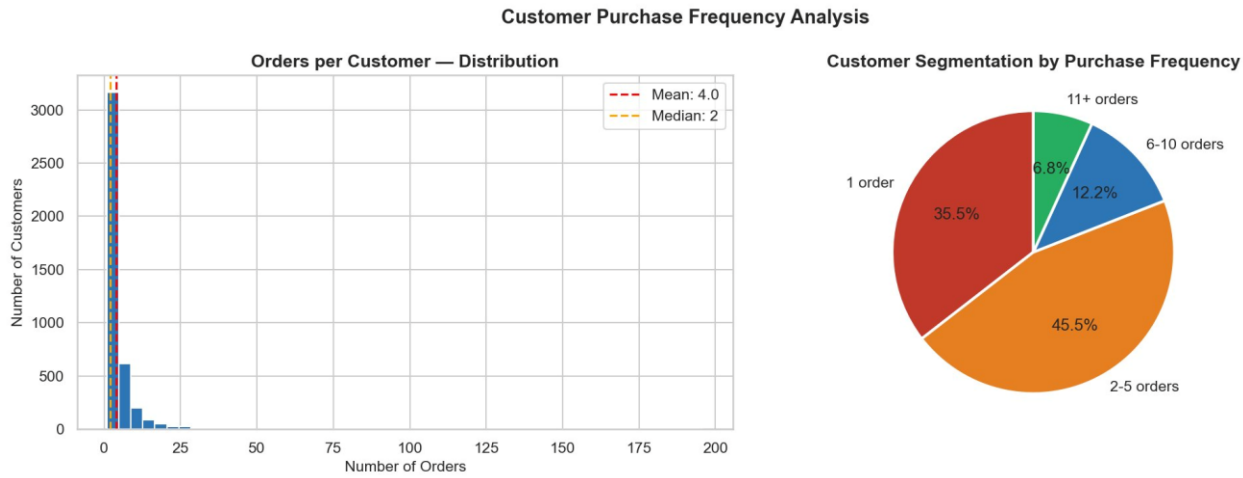

Figure 9 - Customer Purchase Frequency Analysis — Orders per Customer Distribution and Segmentation

Table 4 - Customer Purchase Frequency Analysi

1 order only	35.5% of customers — one-time or lapsed buyers
2–5 orders	45.5% of customers — core repeat segment
6–10 orders	12.2% of customers — loyal mid-tier
11+ orders	6.8% of customers — VIP/high-frequency segment
Mean orders	4.0 orders per customer
Median orders	2 orders per customer

The large gap between mean (4.0) and median (2) confirms extreme right skew — a small number of highly frequent customers pull the mean upward significantly. The 35.5% one-time buyer segment represents a major reactivation opportunity, while the 6.8% VIP segment is the most commercially critical group to retain.

6.3 RFM Distributions

Figure 10 presents the Recency, Frequency, and Monetary distributions derived from the RFM (Recency–Frequency–Monetary) framework — the foundation for customer segmentation.

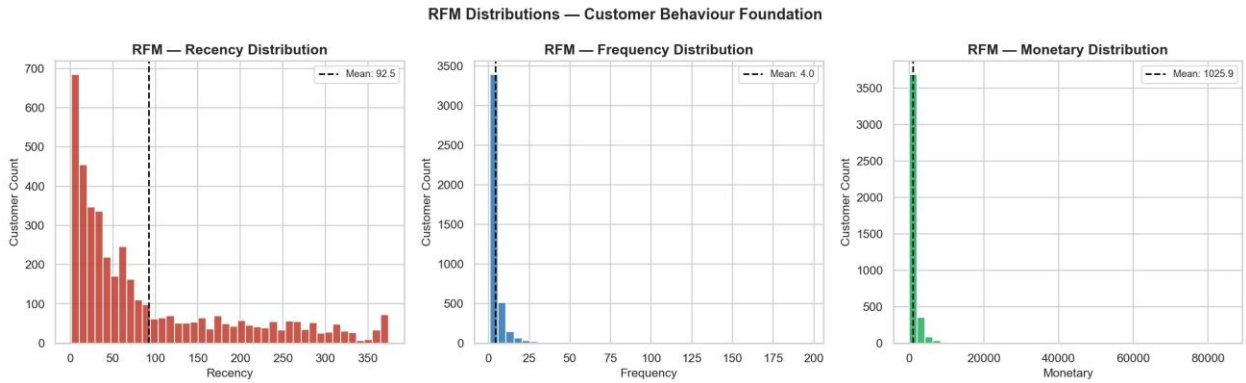

Figure 10 - RFM Distributions — Recency, Frequency, and Monetary Value per Customer

Table 5 - RFM Distributions data

Recency (mean)	92.5 days since last purchase
Frequency (mean)	4.0 orders per customer
Monetary (mean)	£1,025.90 total spend per customer
All metrics	Right-skewed distributions — small VIP tail

The recency distribution shows a heavy concentration of customers who purchased within the last 50 days, with a long tail of lapsed customers (100–365 days). The frequency and monetary distributions both confirm a small high-value VIP segment coexisting with a large base of low-engagement customers. These three distributions directly inform the clustering component of the group project by defining the segmentation axes.

7. Geographic Analysis

7.1 Revenue by Country

Figure 11 presents revenue by country across two panels: all markets (left) and international markets excluding the UK (right), allowing the scale of UK dominance and international opportunity to be assessed simultaneously.

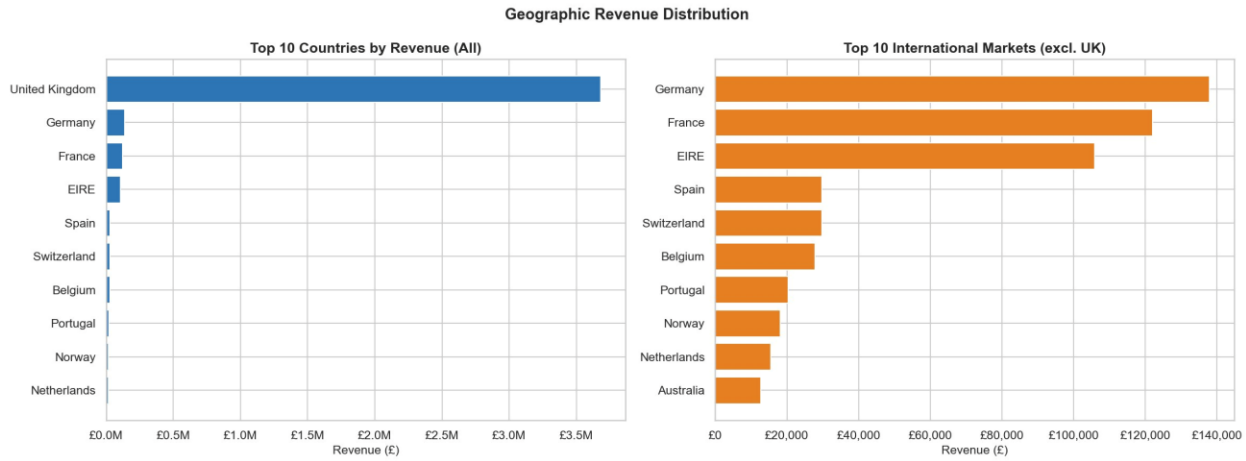

Figure 11 -Geographic Revenue Distribution — All Markets (left) and Top 10 International Markets excl. UK (right)

7.2 Market Share Breakdown

Figure 12 presents the proportional revenue split by country as a pie chart, making the UK's dominance visually unambiguous.

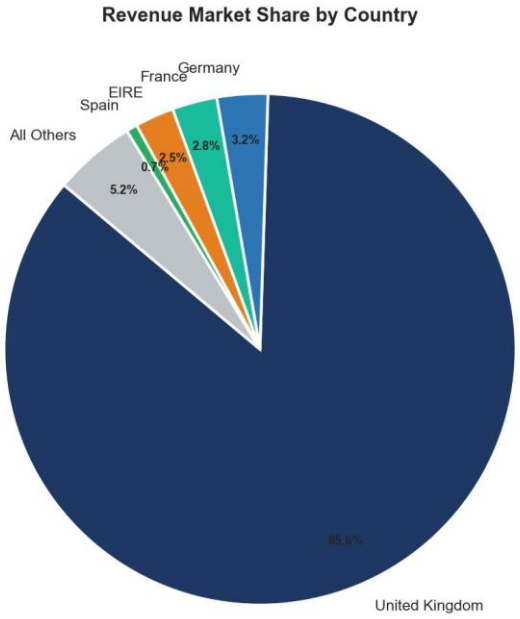

Figure 12 - Revenue Market Share by Country — UK vs International

Table 6 - Revenue Market Share by Country data

United Kingdom	85.6% of total revenue
Germany	3.2% — top international market (~£140K)
France	2.8% (~£120K)
EIRE (Ireland)	2.5% (~£105K)
Spain	0.70%
All Others	5.2% spread across 32 countries

Geographic Concentration Risk

The 85.6% UK revenue concentration represents both a core strength and a strategic vulnerability. A disruption to UK operations or demand would disproportionately impact total revenue. Germany, France, and EIRE together represent approximately 8.5% — a meaningful foundation for international diversification.

8. Correlation & Relationship Analysis

8.1 Correlation Heatmap

Figure 13 presents a lower-triangle Pearson correlation matrix for the five key numerical features in the dataset. Only the lower triangle is shown to avoid redundancy.

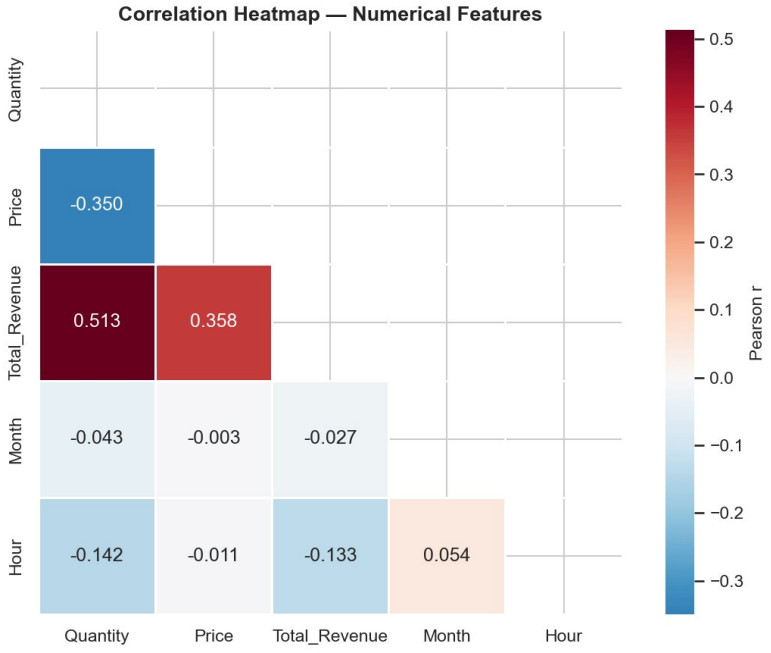

Figure 13 - Correlation Heatmap — Numerical Features (Pearson r)

Table 7 - Correlation Heatmap data

Quantity ↔ Total_Revenue	r = 0.513 — moderate positive correlation
Price ↔ Total_Revenue	r = 0.358 — weak-to-moderate positive
Quantity ↔ Price	r = −0.350 — weak negative (higher price → lower qty)
Month ↔ Total_Revenue	r = −0.027 — negligible
Hour ↔ Total_Revenue	r = −0.133 — negligible

The moderate positive correlation between Quantity and Total_Revenue (r = 0.513) is the strongest relationship in the dataset, confirming that volume is the primary driver of revenue. The negative Quantity–Price relationship (r = −0.350) reflects natural price elasticity: higher-priced items are typically purchased in smaller quantities. Month and Hour show negligible linear correlations with revenue individually, though their combined temporal effect (Q4 surge) emerges clearly in the time-series analysis.

8.2 Scatter Plot Analysis — Revenue Drivers

Figure 14 visualises the Quantity–Revenue and Price–Revenue relationships as scatter plots on a 5,000-record sample, confirming the heatmap findings visually.

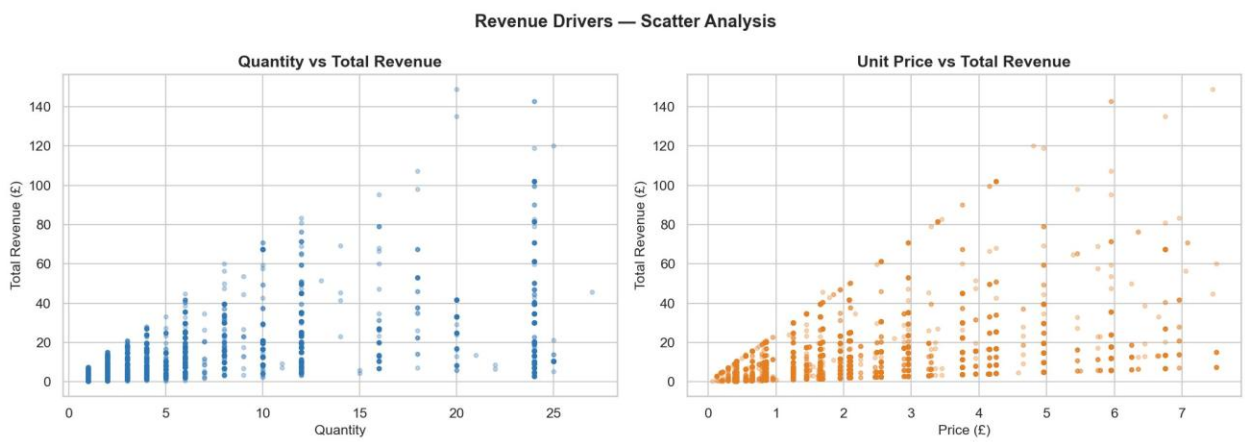

Figure 14 - Revenue Drivers — Scatter Analysis: Quantity vs Revenue (left) and Unit Price vs Revenue (right)

The Quantity–Revenue scatter shows a clear upward trend with increasing variance at higher quantities — bulk orders span a wide revenue range depending on unit price. The Price–Revenue scatter shows a cone-shaped positive relationship: as unit price increases, the range of achievable revenue values widens, reflecting the greater variability in quantities purchased at higher price points.

8.3 Pairplot — Multivariate Relationships

Figure 15 presents a pairplot of Quantity, Price, and Total_Revenue using kernel density estimates on the diagonal and scatter plots on the off-diagonal panels.

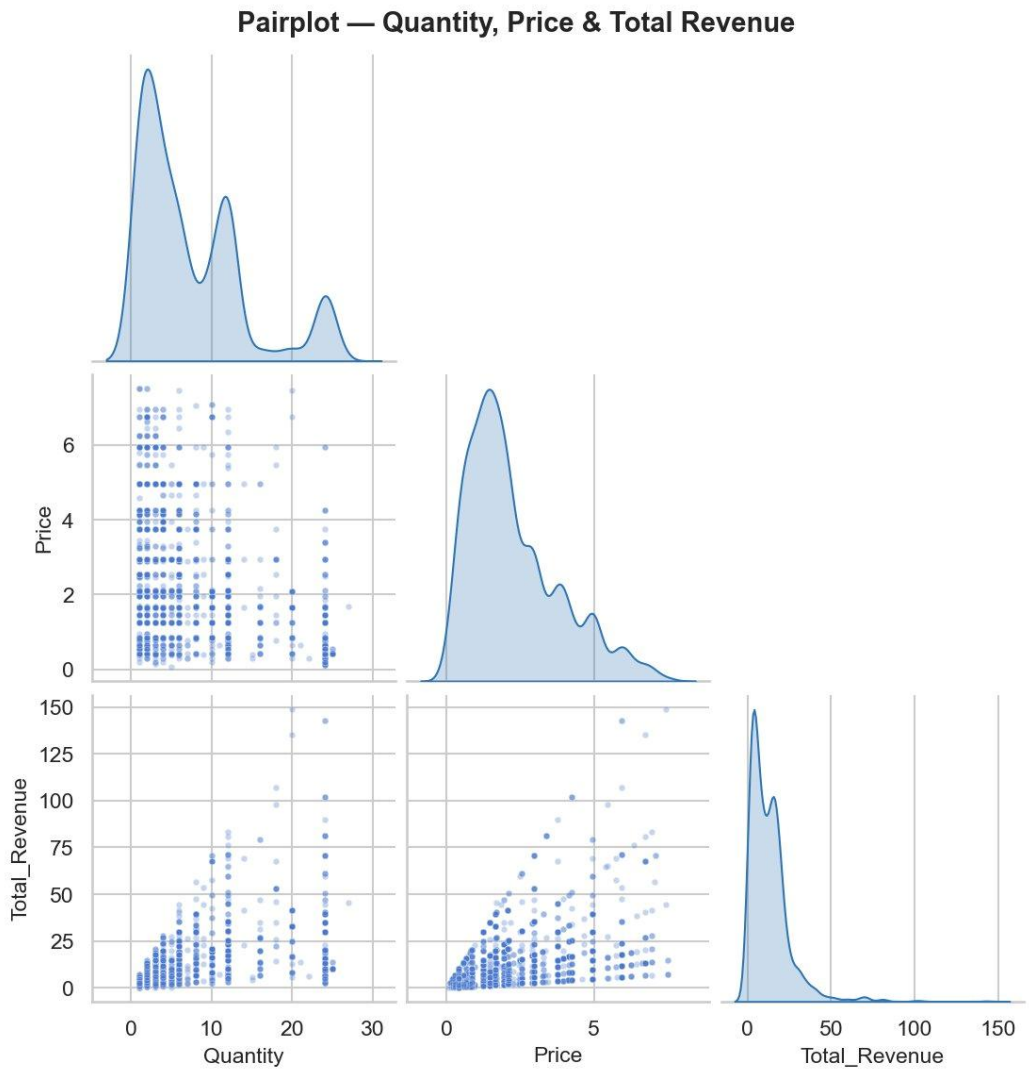

Figure 15 - Pairplot — Quantity, Price and Total Revenue with KDE Diagonals

The pairplot confirms all three distributions are right-skewed (steep KDE peaks concentrated near zero with long right tails). The Quantity diagonal reveals a bimodal distribution — a common pattern for products sold in fixed pack sizes (e.g. packs of 6, 12, 24). The Price diagonal shows a heavily concentrated distribution below £4 with occasional high-price outliers. No strong linear relationship is visible between Price and Quantity, consistent with the r = −0.35 finding.

9. Anomaly Detection & Pareto Analysis

9.1 High-Value Transaction Anomalies

Figure 16 plots all transactions over time, distinguishing normal transactions (blue) from anomalous high-value transactions (red) using the Mean + 3 standard deviations threshold.

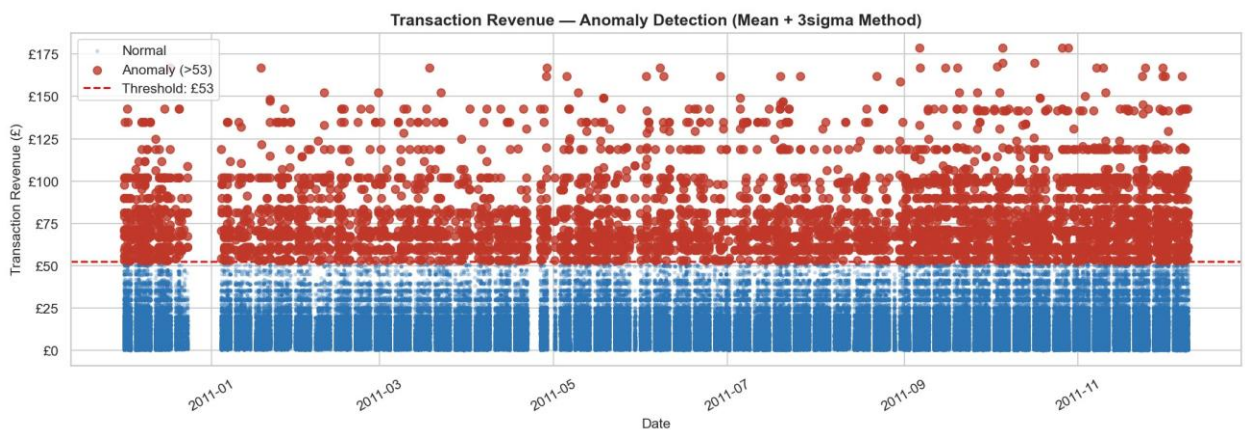

Figure 16 -Transaction Revenue — Anomaly Detection (Mean + 3σ Method)

Table 8 -Transaction Revenue data

Anomaly Threshold	£53.00 (Mean + 3σ)
Anomalous Transactions	~2% of total dataset
Distribution	Spread evenly across the year with Q4 concentration
Likely Explanation	B2B bulk orders from wholesale accounts

Anomalous transactions are not confined to a single time period — they appear throughout the year, with a visible increase in density during Q4. This persistent pattern suggests that high-value transactions represent a distinct segment of wholesale/bulk buyers rather than data errors. These transactions merit separate modelling treatment in the forecasting component.

9.2 Pareto Analysis — Revenue Concentration

Figure 17 presents a Pareto (cumulative revenue concentration) curve showing what percentage of cumulative revenue is attributable to each percentile of customers, ranked highest to lowest.

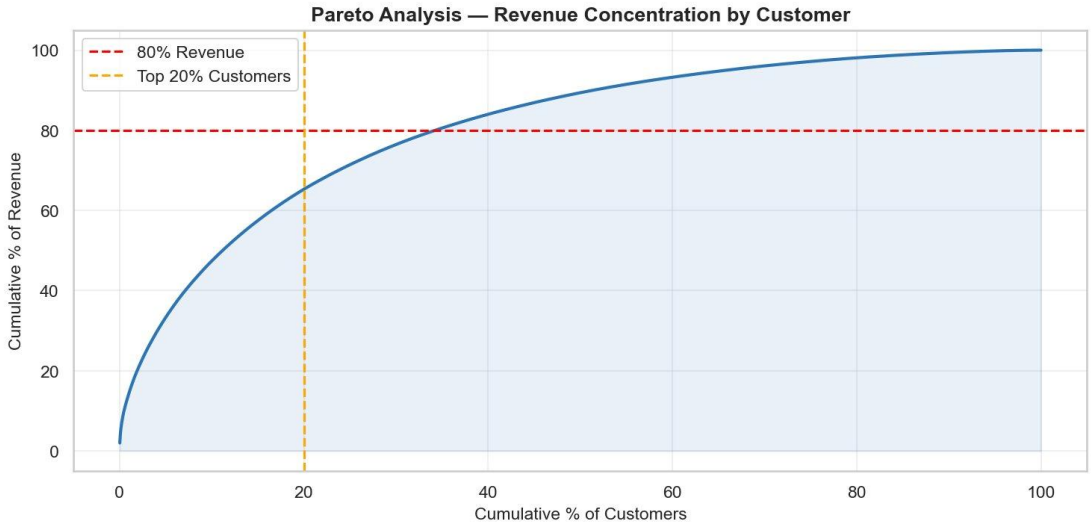

Figure 17 - Pareto Analysis — Revenue Concentration by Customer (Cumulative %)

20% Top 20% of Customers → ~65% of Revenue

The Pareto curve crosses 65% cumulative revenue at the 20th customer percentile — confirming a more concentrated distribution than the classic 80/20 rule. The bottom 50% of customers contribute less than 15% of total revenue.

- The steepness of the curve in the first decile reflects the outsized impact of Customer 14911 and a handful of other VIP accounts.
- The relatively gradual tail beyond the 30th percentile suggests a broad base of low-value customers that are individually insignificant but collectively material.
- This finding directly informs CRM strategy: tiered service models, VIP retention investment, and reactivation campaigns are all data-justified.

10. EDA Summary Dashboard

Figure 18 is the composite EDA summary dashboard, integrating all six core analytical dimensions into a single reference visualisation. It serves as the primary presentation aid and enables cross-dimension pattern recognition at a glance.

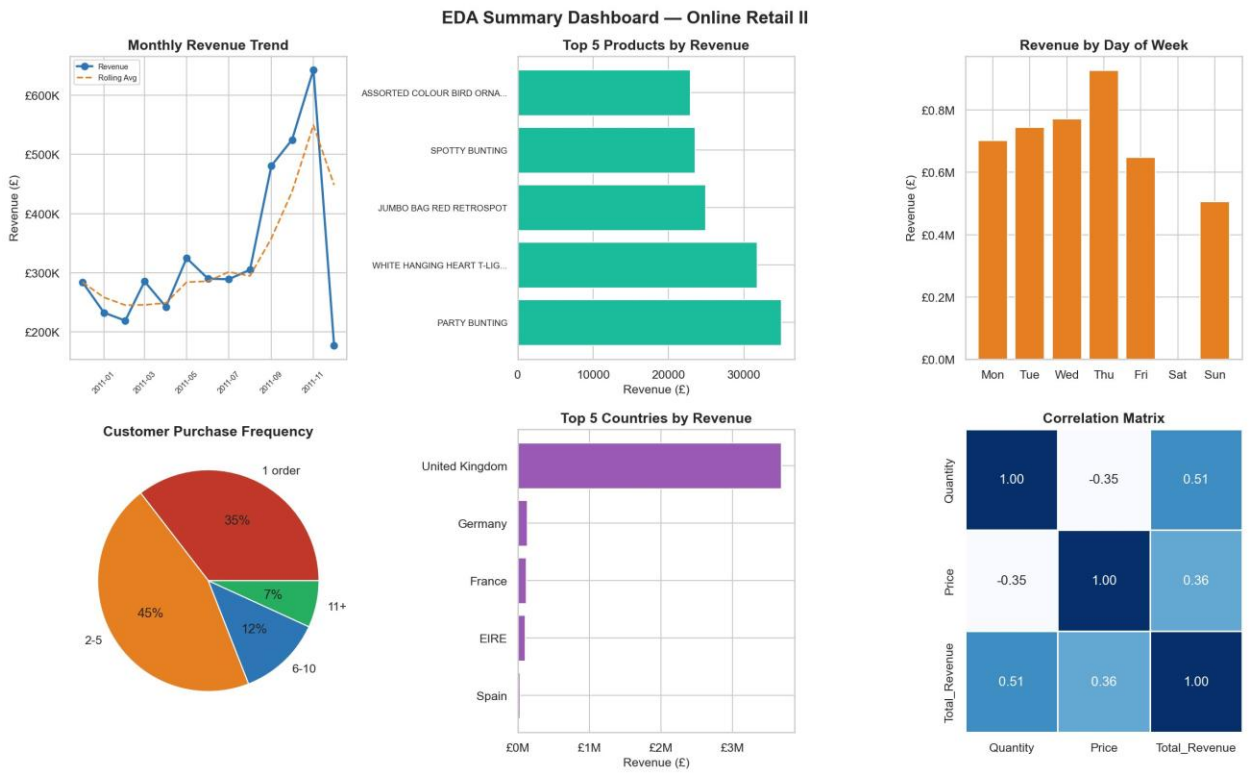

Figure 18 - EDA Summary Dashboard — Online Retail II (All 6 Key Dimensions)

The dashboard panels from left to right and top to bottom are:

- Monthly Revenue Trend with 3-month rolling average — Q4 growth cycle
- Top 5 Products by Revenue — PARTY BUNTING and WHITE HANGING HEART T-LIGHT HOLDER lead
- Revenue by Day of Week — Thursday peak, Saturday zero
- Customer Purchase Frequency — segmented pie chart
- Top 5 Countries by Revenue — UK dominance visualised
- Correlation Matrix (3×3) — Quantity, Price, Total_Revenue relationships

11. Key Insights & Business Recommendations

Q4 Peak Demand

Revenue spikes sharply from September to November each year. Marketing spend, stock procurement, and warehouse capacity should be concentrated in Q3–Q4 to maximise ROI during the peak trading window.

VIP Customer Retention

The top 20% of customers drive ~65% of revenue. A dedicated VIP retention programme — personalised offers, priority fulfilment, loyalty rewards — is directly justified by the Pareto analysis. Customer 14911 alone warrants a dedicated account manager.

International Diversification

UK revenue concentration at 85.6% represents a concentration risk. Germany (3.2%), France (2.8%), and EIRE (2.5%) have demonstrated sufficient purchase behaviour to justify targeted localisation and international marketing investment.

Peak Hour Scheduling

Thursday at 12:00 noon is the single highest-value trading window. Email campaigns, promotions, and flash sales should be timed to land before this peak to capture maximum active engagement.

Product Portfolio Optimisation

The £1–3 price band is the commercial core — stock depth and availability in this band must be protected. PARTY BUNTING leads by revenue and should receive prioritised promotion. Bundling low-price high-volume items with mid-range products could lift average order value.

B2B Segment Investigation

Zero Saturday activity combined with persistent high-value anomalous transactions above the £53 threshold together point to a distinct B2B wholesale customer segment. Separating B2B and B2C cohorts for independent analysis and forecasting would improve model accuracy.

12. Methodology & Technical Notes

12.1 Tools & Libraries

Table 9 - Tools & Libraries

Language	Python 3
Core Libraries	pandas, numpy, matplotlib, seaborn
Notebook	02_eda.ipynb (Jupyter)
Chart Style	Custom palette — #2E75B6, #1ABC9C, #E67E22
Statistical Tools	Pearson correlation, IQR, Mean + 3σ anomaly detection, RFM
Feature Engineering	DayOfWeek, Hour, Month, Year, Price_Band (5 bins)

12.2 Analysis Design Decisions

- A 5,000-record sample was used for scatter plots and pairplots to reduce visual overplotting while preserving representativeness.
- The 3-month rolling average was applied to the monthly revenue series to reduce noise from month-to-month variance.
- The anomaly threshold (Mean + 3σ) was chosen as a conservative, statistically grounded definition that captures genuine outliers without over-flagging normal transactions.
- RFM reference date was set to one day after the maximum invoice date to ensure all customers have a non-zero recency score.
- The correlation heatmap uses the lower triangle only to avoid redundant symmetric display.

12.3 Limitations

- Analysis is confined to the 2010–2011 period; findings may not generalise to current retail conditions.
- Missing CustomerID records (guest/anonymous transactions) were excluded from customer-level analyses, which may undercount the true customer base.
- B2B and B2C transaction types are not explicitly labelled in the dataset; B2B inferences are based on behavioural patterns only.
- The December 2011 revenue dip reflects partial-month data and should not be interpreted as a structural decline.

13. Conclusion

The EDA successfully identified and documented the key operational, commercial, and behavioural patterns in the Online Retail II dataset across eight analytical dimensions. Eighteen visualisations were generated, each paired with statistical interpretation and business-grounded analysis.

The findings establish a robust empirical foundation for the downstream components of the group project. The identification of key segmentation variables (customer value distribution, geographic concentration, product price band, temporal demand patterns) directly informs the K-Means clustering approach. The Q4 demand cycle, customer purchase frequency distribution, and anomaly patterns together define the target variables and feature candidates for the forecasting model.

All code is documented and reproducible in 02_eda.ipynb. All 18 figures in this report are the original outputs of that notebook, embedded in section order to support analysis and findings as required.